Pathway attribution with TabDDPM and a meta-learner

1 Pathway

The comparison is not one batch against another. The anchor is a peak day. The next seven days are seven forward steps. On each step a TabDDPM fit only on the peak imputes one feature block, conditional on the other columns of that day's row. The diffusion class is `GaussianMultinomialDiffusion` from `yandex-research/tab-ddpm` (commit `b476257`). Numerical columns use the Gaussian head. Known columns are reset to the anchor noising process at every reverse step, the RePaint conditioning step. The label passed to the denoiser is drawn from the peak-day label distribution.

The distance estimator is a logistic meta-learner. PO-risk on day t is

$$\mathbb{E}\big[(\mu_t(X_t) - \mu_0(X_t))^2\big],$$

with μ_0 fit on the peak and μ_t fit on day t. Predictive loss is the Brier score of μ_0 transferred onto day t. On the synthetic panel a ridge fit on the peak also reports the latent MSE. Replacing block b by its imputed draw changes each loss by Δ_b. The rank share is $|\Delta_b| / \sum_{b'} |\Delta_{b'}|$. MMD compares the block on day t with the same block on the peak. HSIC compares the block with the day's label.

The last day reports a bootstrap percentile interval for each Δ and a 40-permutation p-value for MMD and HSIC. Across the seven steps, a sign-flip test asks whether the block with the largest mean absolute movement exceeds each other block.

2 Synthetic peak

Day 0 is standard normal, with four blocks of four columns: `stable`, `context`, `driver`, and `noise`. The outcome uses `stable` and `driver`. Over the next seven days the `driver` mean moves by up to 1.5 and its outcome coefficient rises, while `context` receives a smaller location shift and no outcome role. The unconditional TabDDPM sample has MMD 0.0002 against the peak day.

PO-risk rises from 0.005 to 0.050 and latent MSE from 0.337 to 4.496. By day 7 the `driver` share is 0.910 of the PO-risk movement, 0.953 of the Brier movement, 0.940 of the MSE movement, and 0.834 of the MMD. Its PO interval is $[-0.118, -0.059]$ and its Brier interval is $[-0.140, -0.124]$; both exclude zero. The sign-flip p-values of `driver` against the other three blocks are at most 0.028 for PO-risk, Brier, and MSE. Context is the only other block with a significant MMD ($p = 0.024$), and its Brier movement stays near 0.002. HSIC deltas are all below 0.001 and do not rank the blocks. Table 1–3 record the path, the peak day, and the sign-flip tests.

3 IEEE-CIS week after the fraud-rate peak

The local fraud-rate peak with seven usable days after it is transaction day 173 (fraud rate 0.0495). The pathway runs through day 180 (0.0374). Six blocks are used: amount, card, address, counts, timedelta, and hour. The monotone clock is not a feature. The unconditional sample is farther from the peak day here (MMD 0.215) than on the Gaussian panel.

Table 1: Synthetic pathway. The anchor is day 0. Each step is one day closer to the drift peak. Shares are the block's fraction of the total absolute movement.

step	PO-risk	Brier	MSE	driver PO	driver Brier	driver MSE	driver MMD	context MMD
1	0.005	0.236	0.337	0.251	0.441	0.489	0.863	0.000
2	0.014	0.231	0.403	0.794	0.664	0.531	0.886	0.076
3	0.017	0.212	0.486	0.819	0.645	0.678	0.905	0.095
4	0.026	0.195	0.901	0.837	0.871	0.809	0.920	0.071
5	0.031	0.182	1.427	0.328	0.819	0.822	0.864	0.136
6	0.032	0.158	2.589	0.910	0.908	0.920	0.852	0.144
7	0.050	0.122	4.496	0.910	0.953	0.940	0.834	0.166

Table 2: Synthetic day 7, the drift peak. Deltas are the change in the loss when that block is replaced by the TabDDPM draw from the anchor conditional. Intervals are bootstrap percentiles. MMD and HSIC p-values use 40 permutations.

block	PO Δ	PO interval	Brier Δ	Brier interval	MSE Δ	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
stable	-0.003	[-0.011, 0.003]	-0.003	[-0.008, 0.002]	-0.268	0.000	1.000	-0.0004	0.585	0.023	0.025
context	-0.005	[-0.017, 0.002]	-0.002	[-0.003, -0.000]	-0.035	0.084	0.024	-0.0002	0.341	0.043	0.012
driver	-0.100	[-0.118, -0.059]	-0.131	[-0.140, -0.124]	-4.785	0.420	0.024	0.0003	0.049	0.910	0.953
noise	0.003	[-0.005, 0.006]	-0.001	[-0.003, 0.001]	0.000	0.000	1.000	-0.0002	0.024	0.024	0.009

No block has a significant MMD on day 180. The sign-flip test does not separate PO-risk among amount, card, address, counts, and timedelta. The transferred Brier movement is largest for card, and the sign-flip p-value against each other block is at most 0.016, while card's own MMD p-value is 1. Over this week the card block moves the fraud model's predictions, and the covariate blocks do not separate. Tables 4 and 5 give the day-180 numbers and the seven-step tests.

Table 3: Sign-flip test over the seven synthetic steps. The contrast is the absolute movement of the driver minus the other block.

loss	comparison	mean gap	p
PO-risk	driver > stable	0.030	0.008
PO-risk	driver > context	0.028	0.014
PO-risk	driver > noise	0.029	0.028
Brier	driver > stable	0.050	0.014
Brier	driver > context	0.060	0.008
Brier	driver > noise	0.059	0.008
MSE	driver > stable	1.487	0.014
MSE	driver > context	1.803	0.008
MSE	driver > noise	1.810	0.008

Table 4: IEEE-CIS transaction day 180, seven days after the fraud-rate peak on day 173. Same columns as the synthetic peak day. No latent MSE on this table.

block	PO Δ	PO interval	Brier Δ	Brier interval	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
amount	-0.022	[-0.032, 0.002]	0.002	[0.001, 0.005]	0.002	0.098	-0.0001	0.122	0.144	0.009
card	0.012	[-0.014, 0.018]	-0.080	[-0.105, -0.062]	0.001	1.000	-0.0003	0.098	0.077	0.338
address	-0.058	[-0.082, -0.039]	-0.048	[-0.068, -0.033]	0.000	1.000	0.0000	0.073	0.378	0.203
counts	-0.052	[-0.055, 0.006]	-0.051	[-0.063, -0.039]	0.000	0.146	-0.0000	0.122	0.338	0.214
timedelta	0.008	[-0.130, 0.023]	0.029	[0.013, 0.040]	0.000	1.000	-0.0001	0.049	0.055	0.120
hour	0.001	[-0.030, 0.005]	-0.028	[-0.039, -0.016]	0.002	0.073	0.0002	0.024	0.010	0.116

Table 5: Sign-flip test over the seven IEEE-CIS steps after the peak. PO-risk does not separate the blocks. The transferred Brier movement is largest for the card block.

loss	comparison	mean gap	p
PO-risk	timedelta > amount	0.007	0.491
PO-risk	timedelta > card	0.020	0.227
PO-risk	timedelta > address	0.008	0.432
PO-risk	timedelta > counts	0.009	0.423
PO-risk	timedelta > hour	0.029	0.024
Brier	card > amount	0.075	0.008
Brier	card > address	0.019	0.008
Brier	card > counts	0.015	0.016
Brier	card > timedelta	0.054	0.008
Brier	card > hour	0.059	0.008